Retail Inventory & Supply Chain Analysis

An end-to-end data analytics portfolio project

PROJECT SNAPSHOT	DETAIL
Dataset	35,000 records • 25 columns
Tools	Excel • Python • PostgreSQL • Power BI
Analysis areas	Sales, profit estimate, inventory, suppliers, delivery, returns, forecast, channels, warehouses

Purpose

This report summarizes data preparation, analytical approach, key findings, and practical recommendations. It accompanies the project's SQL work and Power BI dashboard.

1. Executive Summary

The project analyzed 35,000 retail inventory and supply chain records to understand sales performance, product-category contribution, replenishment needs, delivery outcomes, returns, demand forecasting, suppliers, sales channels, and warehouse distribution.

Key KPI	Result	Interpretation
Total records	35,000	Records available after cleaning and validation
Total units sold	1,598,379	Total sales volume represented
Revenue	₹1,783,830,481 (₹178.38 Cr)	Units sold × unit price
Estimated gross profit	₹873,544,471 (₹87.35 Cr)	Units sold × (unit price − unit cost)
Reorder required	10,606 records	Stock on hand below reorder level
On-time delivery	22,957 (65.59%)	Largest delivery-status group
Returned units / rate	13,443 / approx. 0.84%	Unit-based return rate

Key signals include a substantial replenishment review list, most records marked On Time, revenue concentrated in Home & Living and Electronics, and nearly half of records classified as Below Forecast. These are leads for investigation, not proof of root causes.

2. Project Objectives

- Measure sales volume, revenue, and estimated gross profit.
- Compare category-level revenue and profitability.
- Identify records that may need inventory replenishment.
- Review delivery outcomes, returns, and forecast status.
- Compare supplier, sales-channel, and warehouse performance.
- Build a Power BI dashboard for business-facing summary and exploration.

3. Tools and Methodology

Stage	Work completed	Purpose
Excel	Initial inspection, duplicate removal, review of blanks and inconsistent values	Improve source consistency
Python / Pandas	Further cleaning, data-type checks, missing-value checks, analytical summaries	Prepare analysis-ready data

PostgreSQL	Loaded cleaned CSV to `retail_inventory`; validated row counts and unique record IDs; ran 10 business analyses	Answer repeatable business questions
Power BI	Connected to PostgreSQL, created DAX measures, KPI cards and charts	Present key results visually

The cleaned dataset contained 35,000 rows and 25 columns. Initial Excel cleaning removed 75 duplicate records. Python checks found 307 missing numeric cells across five fields; these were retained as missing rather than filled with invented values.

4. Data Quality Notes

Field	Missing values
Stock_On_Hand	64
Order_Quantity	66
Shipping_Cost_INR	53
Returned_Units	57
Demand_Forecast_Units	67
Total missing numeric cells	307

- PostgreSQL validation confirmed 35,000 records and 35,000 unique record_id values.
- Metrics involving missing fields may have different effective coverage.
- Revenue and estimated gross profit are project calculations, not audited accounting figures.

5. Key Findings

5.1 Sales and Revenue

Total sales volume was 1,598,379 units. Calculated revenue was ₹1.78 billion (₹178.38 crore), and estimated gross profit was ₹873.54 million (₹87.35 crore). Monthly revenue peaked in August 2025 at ₹158,782,480 and was lowest in September 2025 at ₹140,371,731.

Metric	Result
Highest-revenue month	August 2025 — ₹158,782,480
Lowest-revenue month	September 2025 — ₹140,371,731
Revenue formula	Units_Sold × Unit_Price_INR
Estimated gross profit	Units_Sold × (Unit_Price_INR − Unit_Cost_INR)

5.2 Product Category Performance

Category	Revenue
Home & Living	₹450,457,766
Electronics	₹433,656,364
Furniture	₹385,831,684
Stationery	₹13,009,625

Home & Living generated the highest reported category revenue, followed by Electronics and Furniture. Stationery had the lowest revenue. Revenue should be assessed alongside margin, demand, and stock availability.

5.3 Inventory and Replenishment

Stock status	Records	Share of all records
Reorder required	10,606	30.30%
Stock sufficient	24,330	69.51%
Stock information missing	64	0.18%
Total	35,000	100.00%

About 30.3% of records with non-missing stock values were below reorder level (10,606 of 34,936). These are records requiring review, not necessarily 10,606 distinct SKUs.

5.4 Delivery Performance

Delivery status	Records	Share
On Time	22,957	65.59%
Delayed	6,714	19.18%
In Transit	3,864	11.04%
Cancelled	1,465	4.19%
Total	35,000	100.00%

On Time was the largest delivery-status group. Delayed records accounted for 19.18% of all records. Further breakdowns by supplier, warehouse, carrier, and destination could help identify where operational follow-up is useful.

5.5 Supplier Performance

Observation	Result
Highest supplier revenue	Metro Wholesale — ₹233,782,350
Metro Wholesale average lead time	6.57 days
Highest units sold among noted suppliers	Brightline Imports — 203,817 units

Supplier selection should consider lead time, delivery reliability, returns, product availability, and cost—not revenue alone.

5.6 Returns

Observation	Result
Total returned units	13,443
Total units sold	1,598,379
Overall unit return rate	Approximately 0.84%
Highest category return rates	Pet Supplies and Grocery — approximately 0.89% each
Most returned units	Home & Living — 3,876
Lowest category return rate reported	Furniture — approximately 0.76%

The overall rate is returned units divided by units sold. It is a unit-based rate, not the percentage of orders returned. Category counts and rates can differ because sales volumes differ.

5.7 Demand Forecast

Forecast comparison	Records	Share of all records
Below Forecast	17,111	48.89%
Above Forecast	13,480	38.51%
Matched	4,342	12.41%
Forecast missing	67	0.19%

Nearly half of records were classified as Below Forecast, while 12.41% were Matched. Validate the forecast definition and compare patterns by category, month, and warehouse before drawing operational conclusions.

5.8 Sales Channel Performance

Sales channel	Revenue
Retail Store	₹363,673,139
Mobile App	₹361,994,517
Wholesale	₹358,329,613
Marketplace	₹357,016,206
Website	₹340,851,655
Unknown	₹1,965,351 (51 records)

Revenue was relatively close across Retail Store, Mobile App, Wholesale, and Marketplace. Unknown channel values should be reviewed to improve attribution and channel comparisons.

5.9 Warehouse Performance

Observation	Result
Highest inventory value reported	Mumbai — ₹533,385,255
Lowest inventory value reported	Kolkata — ₹478,909,750
Highest reorder-required record count	Hyderabad — 1,800 records

Inventory value alone does not measure warehouse efficiency. Consider stock age, demand, stockouts, delivery performance, and inventory turnover as well.

6. SQL Analysis and Power BI Dashboard

Ten business-focused SQL analyses were prepared:

- Overall KPIs; monthly sales and revenue; category sales and profit; supplier performance; inventory reorder status.
- Delivery-status percentages; returns by category; forecast versus actual; sales-channel performance; warehouse performance.

The Power BI report connects to PostgreSQL table `retail_inventory`. It includes KPI cards for Total Revenue, Total Units Sold, Estimated Gross Profit, Reorder Required Records, Inventory Value, and Return Rate, with visuals for monthly revenue, category revenue, inventory status, delivery status, and supplier revenue.

Return Rate should be calculated as DIVIDE(total returned units, total units sold) and formatted as Percentage in Power BI. Do not multiply by 100 in the DAX measure if Percentage formatting is applied. Based on 13,443 returned units and 1,598,379 units sold, the expected rate is approximately 0.84%.

7. Business Recommendations

1. **Prioritize replenishment.** Review the 10,606 records below reorder level, prioritizing items with stronger demand, long lead times, or high stockout impact.
2. **Investigate delays.** Break delayed records down by supplier, warehouse, and destination; compare lead times and cancellations before assigning causes.
3. **Review forecast gaps.** Analyze Below Forecast and Above Forecast patterns by category and month, and validate forecast definitions and time periods.
4. **Protect category availability.** Monitor Home & Living and Electronics because they contribute the largest reported category revenue; pair revenue with margin and stock availability.
5. **Improve data completeness.** Reduce Unknown channel values and validate missing stock, returned-unit, shipping-cost, and forecast data at source.
6. **Track returns with context.** Monitor both returned-unit counts and return rates; investigate return reasons before changing product or inventory policies.
7. **Use a balanced supplier scorecard.** Compare revenue, lead time, on-time delivery, returns, and unit cost.
8. **Balance warehouse stock using demand.** Review inventory value and reorder counts alongside demand, aging, and service levels before transferring stock.

8. Limitations and Interpretation

- The dataset is synthetic and designed for portfolio practice; do not present findings as actual company performance.
- Revenue is units sold × unit price. Estimated gross profit excludes other expenses such as logistics, returns costs, overhead, taxes, and discounts unless already reflected in the input fields.
- Missing numeric values were retained as missing; analyses may have different effective record coverage.
- A record is not necessarily a distinct SKU, order, shipment, or supplier transaction unless the dataset grain confirms it.
- These summaries do not establish causes; recommendations require validation with operational context.

9. Conclusion

This project demonstrates an end-to-end analytics workflow across Excel, Python, PostgreSQL, and Power BI. It highlights replenishment needs, delivery delays, forecast variance, category revenue concentration, and data-completeness opportunities. The dashboard gives a compact overview while SQL queries support deeper investigation.

Appendix: Core Metric Definitions

Metric	Definition
Total Revenue	SUM(Units_Sold × Unit_Price_INR)
Estimated Gross Profit	SUM(Units_Sold × (Unit_Price_INR – Unit_Cost_INR))
Reorder Required	Count records where Stock_On_Hand < Reorder_Level, excluding missing stock
Unit Return Rate	SUM(Returned_Units) ÷ SUM(Units_Sold); display as percentage
Delivery share	Records in a delivery status ÷ all records
Inventory value	SUM(Stock_On_Hand × Unit_Cost_INR), where stock is available